

Natural Language SQL Agent

for Telecom Business Questions

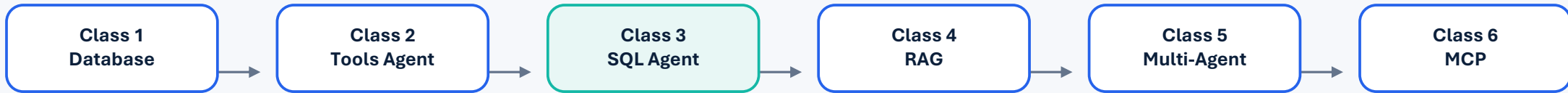
Using the Zain Jordan Customer 360 SQLite Database

**Today we move from
controlled tools to
flexible database
analytics.**

SQL Agent = natural language → query →
insight

Where Class 3 Fits

From database access to AI-powered business intelligence.



What changes today?

Class 2 agents used controlled Python tools. Class 3 agents can inspect the database and generate SQL for wider business questions.

Why it matters

This is the bridge from customer-specific support workflows to management-level analytics, dashboards, and capstone ideas.

Class 3 Learning Outcomes

By the end of this session, participants should be comfortable with the SQL agent workflow.

01

Understand the difference between a tools agent and a SQL agent.

02

Connect LangChain to the SQLite Customer 360 database.

03

Create SQLiteDatabase and SQLiteDatabaseToolkit components.

04

Ask telecom business questions in natural language.

05

Explain SQL agent guardrails and production safety risks.

06

Generate capstone-ready analytics use cases.

Class 2 Recap

The previous class gave the agent controlled tools.

What we built

- Python DB functions
- LangChain tool wrappers
- Customer Care AI Agent
- Retention-focused agent

Best for

Customer-specific workflows:

- Analyze customer 42
- Check churn risk
- Review complaints
- Draft care message

Limitation

The agent can only use predefined tools. It cannot freely explore wider business questions.

TODAY'S UPGRADE

From controlled tool calls to flexible natural-language database analysis.

The SQL agent can inspect schemas, generate SQL, execute queries, and explain the business meaning.

Why a SQL Agent?

Business users ask questions in English, but enterprise data lives in relational databases.

The business problem

Managers often need answers from the database, but they may not know table names, joins, or SQL syntax.

Examples:

- Which cities have high churn?
- Which campaigns converted best?
- Which complaint categories are growing?

The SQL agent solution

The agent translates a natural-language question into a database query, runs the query, and explains the result in business language.

It acts like a lightweight business intelligence assistant.

Tools Agent vs SQL Agent

Both are useful, but they solve different kinds of problems.

Class 2: Tools Agent

Access style

Uses predefined Python functions.

Best for

Controlled customer workflows.

Example

“Analyze customer 42.”

Strength

Safer and easier to control.

Class 3: SQL Agent

Access style

Generates SQL dynamically.

Best for

Flexible analytics questions.

Example

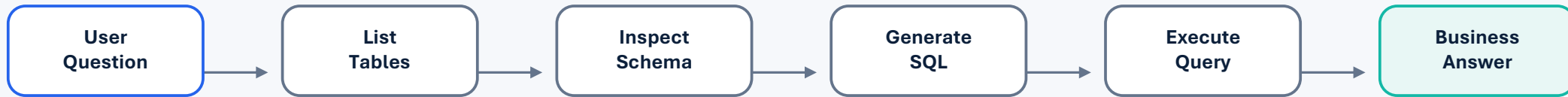
“Which city has the highest churn?”

Strength

More flexible for discovery.

SQL Agent Architecture

The agent follows a controlled sequence to answer a question.



Key idea

The agent does not guess table structure. It uses tools to list tables and inspect relevant schemas before querying.

Trainer emphasis

Show the tool calls in verbose mode first so learners understand the workflow behind the answer.

Zain Jordan Database Areas

The SQL agent can explore multiple telecom business domains.

Customer Core

customers
accounts
subscriptions

Products

plans
addons
subscription_addons

Billing

invoices
payments
transactions

Experience

complaints
support_interactions
customer_satisfaction

Risk & Value

customer_churn_scores
customer_value_segments
customer_monthly_summary

Network & Campaigns

network_events
network_towers
campaigns

This breadth is why the same database works for SQL Agent, RAG, multi-agent, MCP, and capstone projects.

Manual SQL Warm-Up

Start with a normal query before asking the agent to generate one.

BUSINESS QUESTION

Which cities have the most high-risk churn customers?

```
SELECT c.city, COUNT(*) AS high_risk_customers
FROM customers c
JOIN customer_churn_scores ch
  ON c.customer_id = ch.customer_id
WHERE ch.risk_level = 'High'
GROUP BY c.city
ORDER BY high_risk_customers DESC
LIMIT 10;
```

Teaching point: SQL agents are useful, but learners should still understand the query pattern.

Building Blocks of the SQL Agent

Five components create the end-to-end natural language SQL experience.

- | | | |
|----|------------------------|--|
| 01 | SQLite Database | The local Zain Jordan Customer 360 `.db` file. |
| 02 | SQLDatabase | LangChain wrapper that exposes schema and query access. |
| 03 | Chat Model | The LLM that decides next steps and explains results. |
| 04 | SQL Toolkit | Tools for list tables, inspect schema, check query, and execute query. |
| 05 | Agent Prompt | Rules for safety, row limits, SELECT-only behavior, and business interpretation. |

Notebook Flow

How the Class 3 notebook is organized.

Setup

Install libraries, load key, upload DB

SQLDatabase

Connect LangChain to SQLite

Agent

Add safety prompt and run demos

Warm-Up

Manual SQL queries for business context

Toolkit

Create database tools for the agent

Exercises

Business questions and capstone prompts

Core Demo Questions

Start simple, then move into real telecom analytics.

Beginner

- How many customers?
- Customers by city
- Active vs inactive
- Plans by monthly fee
- Churn risk counts

Intermediate

- High churn by city
- Average churn by segment
- ARPU by value segment
- Top complaint categories
- Support channels by volume

Business-ready

- High value + high churn
- Campaign conversion rate
- Network events by impact
- Complaint + churn overlap
- Executive insight summary

Demo Path: From Question to Insight

Use one question to explain the full cycle.

Question

“Which customer value segments have the highest average ARPU and six-month revenue?”



Trainer note

Emphasize that the final answer should include the numbers and a business interpretation, not just a raw table.

SQL Agent Safety

The most important professional habit in this class.

Training environment

- Synthetic data
- Local SQLite file
- Read-only style prompt
- Row limits
- No destructive actions
- Visible agent steps during demo

Production guardrails

- Read-only credentials
- Table and column restrictions
- Sensitive data masking
- Query allowlist or validation
- Logging and monitoring
- Human review for sensitive decisions

Prompt Rules for the SQL Agent

The system prompt controls safety, accuracy, and answer quality.

```
You are a telecom BI SQL agent.
```

```
Rules:
```

- Use SELECT queries only.
- Never modify the database.
- Inspect schemas before querying.
- Limit results unless requested.
- Do not guess missing facts.
- Explain results in business language.

```
Final answer:
```

1. Direct answer
2. Key numbers
3. Business interpretation
4. Recommended next action

Participant Exercises

Move learners from running demos to asking useful questions.

Exercise 1

Simple questions

- How many customers?
- How many plans?
- How many complaints?
- How many network events?

Exercise 2

Grouped analysis

- Customers by city
- Customers by segment
- Complaints by severity
- Support by channel

Exercise 3

Business insight

- Highest churn cities
- Highest ARPU segments
- Best campaigns
- Negative sentiment channels

Common Mistakes to Teach

Make participants better prompt writers and better reviewers.

Vague question “Tell me about customers.”

Too much at once “Analyze everything.”

No row limit “Show all customers.”

Blind trust Not checking tables, filters, or logic.

No interpretation Returning only raw rows without business meaning.

Capstone Project Patterns

Day 3 gives teams flexible analytics capability for their final builds.

Executive KPI Assistant

Revenue, churn, complaints, campaigns, network overview

Complaint Intelligence Assistant

Top categories, severity, sentiment, repeat issues

Network Impact Assistant

Events, affected customers, city and technology impact

Churn Analytics Assistant

Risk segments, high-value churn, city-level patterns

Campaign Performance Assistant

Conversion rate, target segment, offer effectiveness

Revenue Insight Assistant

ARPU, value segments, billing and payment trends

What Class 3 Produces

A working analytics assistant plus capstone-ready thinking.

01

SQLDatabase

02

SQL Toolkit

03

SQL Agent

04

BI Demos

05

Safety

06

Capstone Ideas

End-of-class capability

Participants can ask structured business questions about churn, revenue, complaints, campaigns, support interactions, and network events using natural language. They also understand why guardrails are mandatory before production use.

Closing and Next Step

Class 3 completes the analytics foundation.

Today's milestone

We moved from controlled customer tools to flexible database analytics. The SQL agent can inspect the database, generate SQL, execute queries, and explain business results.

Next class

RAG from telecom database content. We will convert selected rows into searchable knowledge for plan, campaign, support, and customer-care recommendations.

Core message: SQL Agent gives flexible analytics. RAG gives contextual knowledge retrieval.